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Electricity provision and industrial development: Evidence from India ☆

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ABSTRACT

I investigate the effect of electricity provision on industrialization using a panel of Indian states for 1965–1984. To address the endogeneity of investment in electrification, I use the introduction of a new agricultural technology intensive in irrigation (the Green Revolution) as a natural experiment. As electric pumpsets are used to provide farmers with cheap irrigation water, I use the uneven availability of groundwater at the start of the Green Revolution to predict divergence in the expansion of the electricity network and, ultimately, to quantify the effect of electrification on industrial outcomes. I present a series of tests to show that the electrification channel remains the most important one among alternative explanations that could link groundwater availability to industrialization directly or indirectly. Results show that an increase in one standard deviation in the measure of electrification is associated with an increase of around 14% in manufacturing output for a state at the mean of the distribution.

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1. Introduction

The adequate supply of infrastructure goods is increasingly acknowledged as one key factor in generating a conducive environment for industrial and economic development. The World Bank currently directs 35% of their lending portfolio to infrastructure projects with the idea that "infrastructure has a central role in the development agenda and is a major contributor to growth, poverty reduction and achievement of the Millennium Development Goals" (World Bank, 2005). However, average expenditure on infrastructure for developing countries is only around 3% of GDP. Among developing countries, South Asia and Sub-Saharan Africa perform particularly poorly in a range of infrastructure goods, such as water and sanitation, telecoms and electricity access. For example, around 43% of the population has access to the electricity network in Southern Asia, compared to at least 90% in Latin America, Eastern Europe and East Asia.

Access to reliable sources of electrification is, according to World Bank investment climate surveys, one the greatest obstacles to industrial development in India and the rest of Southern Asia. In particular, India's poor infrastructure in general, and the power sector in particular, is seen as one of the reasons behind India's slow export growth during the 1990s, limiting their comparative advantage in labor intensive products (World Bank, 2000). Public agencies, economic journalists and sector analysts[1] are among those who point to the electricity sector's poor performance as heavily affecting the growth and development possibilities of the Indian economy since independence, to the point that "the poor quality of electricity has been the single greatest deterrent to India's economic growth and development' (US DoE, 2003). Yet the history of Indian electrification is not one of uniform failure. For example, around 10% of villages on average were electrified by 1965, with some states like Tamil Nadu close to 50% and others such as Assam or West Bengal less than 3%. In 1984, the average increased to over 75% and some states like Punjab achieved full village electrification.

I use this variation across regions and over time within India to investigate the effect of electricity provision on industrial development, by examining a panel of Indian states between 1965 and 1984. India's federal political organization gives each state full responsibility for creating, expanding and managing the electricity network, from electricity generation to retail. This provides significant variation in the extent of the physical network, allowing me to test whether the gap in infrastructure provision is associated with unequal industrialization levels.

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[1] For example, "(..) electricity is unusable for industry and of such poor quality that power surges wreck equipment." (The Economist, 22/9/2005). See also TERI (1999).

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Assessing and quantifying the impact of investment in electrification on economic outcomes in the long run is a difficult task, since it is hard to determine the underlying driving force. The resulting endogeneity concerns arise because of reverse causality and unobserved state characteristics (such as the business or political climate) that might explain why some states are better prepared to provide a better electricity network than others. In such cases, OLS estimates would be biased. To address this problem, I investigate a source of exogenous variation to predict the expansion of the electricity network in order to estimate its impact on industrial development in an instrumental variable (IV) strategy. In particular, I use the start of the Green Revolution – an agricultural technology intensive in irrigation introduced in India in the mid 60s – as a natural experiment. The successful introduction of the new High Yield Variety (HYV) seeds depended, among other factors, on the provision of timely irrigation. As electric pumpsets were used to provide cheap irrigation water, the uneven availability of groundwater across states – as measured by the thickness of the watertable – was an important determinant of electrification.

I use the time-varying effects of the initial groundwater availability as an instrument for the expansion of the electricity network in the following way. I first document the relationship between groundwater availability and rural electrification (first stage) and manufacturing output (reduced form). I subsequently present the main results, and discuss the most important concern regarding the empirical strategy, namely the validity of the exclusion restriction: groundwater or HYV adoption could affect industrial outcomes through channels other than electrification. For example, greater rural incomes can increase the demand for manufacturing goods, positive shocks to agricultural productivity release cheap labor to be employed in the manufacturing sector or groundwater can be used directly by industries. Similarly, the political features of a state that determine a successful process of HYV adoption and electrification could also explain industrial growth. I present a discussion of alternative channels and a series of checks that suggest that the assumption that the exclusion restriction holds is not implausible. As a consequence, there is room to interpret the IV results as a causal effect of electrification on industrial output.

Results show that an increase in one standard deviation in the measure of electrification is associated with an increase of around 14% in manufacturing output for a state at the mean of the distribution. Electrification is also associated with more factories and output among smaller firms, showing evidence of an extensive margin of expansion. However, these effects are small to account for the increase in manufacturing output. Rather, most of the effect seems to come from the intensive margin of expansion.

This paper contributes to the growing literature on infrastructure, reinforcing the idea that geographic characteristics can be used to explain differential investments in expensive infrastructure projects, and address its endogenous placement. For example, Duflo and Pande (2007) instrument dams placement in India using information on rivers' gradients and Dinkelman (2010) instruments electrification in South Africa with land gradients.

The paper is organized as follows. Section 2 provides some background of the electricity sector in India and its relation to the Green Revolution and presents the data. Section 3 discusses the IV strategy, presenting the first stage and reduced form evidence. I then show OLS and IV results of linking electricity indicators to various manufacturing outcomes. In Section 4 I discuss alternative explanations and In Section 5 I show that results are robust to different measures of groundwater and electrification. Section 6 concludes.

2. Background and data

The empirical contribution of this paper (to document the effect of electrification in industrial development) draws both from the organization of the electricity sector in India and from the uneven introduction of an agricultural technology that increased the demand for powered irrigation. In particular, I argue that groundwater availability has been a major source of electrified irrigation since the start of the Green Revolution and, thus, an important factor in explaining rural electrification across states. I then explore whether this mainly agricultural phenomenon could have benefited untargeted industrial users (in both rural and urban areas) by looking at a panel of 15 Indian states between 1965 and 1984.[2]

The starting point is the Green Revolution, i.e. the introduction of high-yield varieties (HYV) in the mid 1960s, after persistent food shortages in newly independent India. Farming was mainly for subsistence, rainfed and characterized by the use of primitive techniques (Chakravarti, 1972). The Third Five Year Plan laid out by the Planning Commission, and starting in 1961, set ambitious targets in terms of agricultural production so that "the economy will become self-sufficient in the supply of foodgrains". After the failure of some programs such as the Intensive Agricultural District Programme (IADP) and the Intensive Agricultural Areas Programme (IAAP), in 1966–7 the introduction of the High Yielding Variety Seed Programme provided the expected breakthrough, by providing farmers with hybrid seeds scientifically adapted to India's domestic conditions. The Green Revolution – in order to become a 'revolution' – depended on a series of factors, especially the adequate and timely supply of water. It is no surprise that states like Haryana and Punjab – both at the forefront of the Green Revolution – achieved a share of irrigated area around 60% and 90% respectively in the mid-1980s, while other relatively rich states like Gujarat and West Bengal were around the 25% mark. Irrigation could be provided by using canals supplying water from a dam or reservoir or the installation of deep or shallow powered tubewells, i.e. diesel or electricity operated pumps that lift the water from the water table. Bharadway (1990) notes that "the rate of increase of irrigation by wells/tubewells was higher than that by canals, and accelerated remarkably during the period 1969–1980 when there was a spurt in private tubewells, especially in the late sixties". Moreover, according to the National Commission on Agriculture, Ministry of Agriculture and Irrigation of India (1976), electricity-powered pumpsets were the cheapest option for farmers, up to half the cost of operating diesel pumpsets. It follows that places where groundwater was available and abundant were better suited for a cost-effective utilization of electric pumpsets that would provide the timely and precise irrigation needed by HYV seeds.

The data on groundwater availability comes from the Water Resources section of the III Volume of the National Atlas of India, "Climatic and Biogeographical Maps" (NATMO, 1982). The maps are organized in five plates (Northern, Western, Central, Eastern and Southern India) and outline four different categories in terms of groundwater availability, namely (1) fairly extensive thick aquifers occurring beyond 150 m, (2) aquifers with limited extent occurring between 100 and 150 m, (3) aquifers with restricted extent occurring up to 100 m and (4) aquifers with restricted extent occurring at various depths. On the basis of these categories, I create a variable measuring the proportion of state area covered by the deepest aquifers, according to category (1).[3] This measure is the most important for my empirical strategy, since groundwater availability has been the major source of electrified irrigation after the Green Revolution (see McGuirk and Mundlak, 1991, and Foster and Rosenzweig, 2008).[4] The data for adoption of HYV seeds come from the district level dataset "India Agriculture and Climate Dataset", compiled by Sanghi, Kumar, McKinsey, Jr. (1998) for the years 1957–

[2] The data includes all Indian states according to the state configuration in 1966. I only exclude Jammu & Kashmir and smaller Union Territories because there is no consistent data available.

[3] In Section 5 I show that results hold for alternative measures of groundwater, by using categories (2) and (3) aquifers and information on net groundwater availability in 2004 estimated by Central Ground Water Board (2006).

[4] Note that I use a measure of thickness of the water table at the outset of the Green Revolution. Subsequent use (and depletion) of the available groundwater could be endogenous to HYV adoption and electrification.

58 to 1986–87. I use information on the net proportion of area cropped with HYV seeds.

I subsequently investigate the process of electrification and industrial growth across states. There is a constitutional arrangement in India that ensures state independence from the central government in designing electricity policies. The Electricity Supply Act created in 1948 fully vertically integrated State Electricity Boards (SEB), in charge of coordinating electricity generation, transmission and distribution and commercialization at the state level, whose Board members are appointed by the state government (Electricity Supply Act, Chapter III, 5.2). An important aspect of the power sector organization is that during the period analyzed, there was negligible cross-state trade of electricity: on average, 97% of available electricity was generated within each state. The electricity data comes from "Public Electricity Supply — All India Statistics", published annually by the Central Electricity Authority between 1950 and 1985. The measure of rural electrification is the number of agricultural units connected to the electricity network per 1000 people.[5]

Data for industrial development and performance indicators are taken from different publications from the Department of Statistics, Ministry of Planning, Government of India. Manufacturing output comes from "Estimates of State Domestic Product" and measures of the stock of fixed capital, value added and number of factories come from the "Annual Survey of Industries". The data on industrial development is only used until 1984. The main reason is that in this period there was a relatively stable institutional framework for industrial producers: the manufacturing sector was regulated by the Industries Development and Regulations Act that had been in place since 1951. The following year, in 1985, India started a wave of reforms in the manufacturing sector by delicensing around a third of all three-digit industries. From the outset, these reforms have changed considerably the dynamics of the manufacturing sector (see Aghion et al., 2008). However, this twenty-year period from 1965 still provides me with a time span long enough to observe whether there is a persistent divergence in electrification across states that could be associated with initial groundwater after the beginning of the Green Revolution.

There are other state characteristics included as controls, such as population density, proportion of rural and literate population, mean household rural expenditure, credit, education and development expenditure and political outcomes.[6] Table 1 provides summary statistics for the main variables used in this paper.

The evolution of electrification and manufacturing output and their link to the previously-defined measure of initial groundwater availability is provided in Table 2. The table shows significant variation across states in the measure of groundwater, initial levels of electrification and industrialization and subsequent changes. Fig. 1 shows a virtually flat relationship between rural electrification and manufacturing output in 1965. Fig. 2 shows a positive correlation between changes in agricultural electrification and manufacturing output in the subsequent twenty years. Some elements of this relation, I argue, can be explained by the initial endowments of groundwater that triggered a differential process of electrification across states.

3. Baseline results

In this section I explore more systematically whether initial groundwater can be associated with an expansion of the electricity network (the first stage in the IV strategy) and with increases in manufacturing outcomes (the reduced form results). Subsequently, I present and discuss the OLS and IV results, noting that the exclusion restriction assumption is open to criticism. I explore its plausibility in the following section.

Table 1
Summary statistics.

Variables	Mean	Std. dev.	Min	Max
Agricultural connections (per 1000 people)	5.11	5.45	0	23.24
Log manufacturing output per capita	7.07	0.54	5.98	8.30
Groundwater (share of area)	0.20	0.26	0	0.8
HYV adoption (share of cropped area)	0.23	0.17	0	0.92
Rural population (share)	0.79	0.07	0.64	0.93
Population density	0.26	0.15	0.06	0.68
Literacy (share of population)	0.39	0.13	0.19	0.82
Log education expenditure per capita	−7.23	0.83	−9.05	−5.12
Log development expenditure per capita	−6.15	0.76	−8.05	−4.12
Log mean rural household expenditure	4.00	0.17	3.58	4.49
Log total credit per capita	4.90	0.92	2.49	6.76
Log non-agricultural credit per capita	4.47	0.99	2.22	6.63
Congress Party Chief Minister (share)	0.69	0.46	0	1
Congress Party vote (share)	0.39	0.11	0.09	0.56

3.1. Groundwater as an instrument

I start by providing evidence of the link between groundwater availability and the diffusion of HYV seeds. This helps to back the claim that states followed either a path of progressive adoption of the new agricultural technology that created a demand for rural electricity, or a path of traditional farming. I then present evidence towards the validation of the first stage and reduced form results.

3.1.1. Groundwater and HYV adoption

The adoption of HYV seeds in rural India has been subject to a number of studies. As noted by Sharma and Dak (1989) or Kohli and Singh (1997), among others, there are many institutional factors that may claim a share of responsibility in the adoption of this new technology (e.g. price incentives, land distribution or human capital.) However, since these variables might be correlated with (or determined by) industrial output there is a need for exogenous source of variation that explains HYV successful adoption. The answer lies in geographic and climatic characteristics. Foster and Rosenzweig (1996) point out that "a feature of the Green Revolution in India is that the ability to exploit the new seeds profitably was substantially different across India because of exogenous differentials in local soil and weather conditions". Similarly, Evenson and McKinsey (1999) show that climatic and soil conditions resulted in the diffusion of HYV seeds.

In this section I further investigate these claims by looking at whether groundwater availability can be associated to the evolution of HYV adoption, a necessary step to explain subsequent electrification. To that end, I first run a regression of the form

$$HYV_{st} = \beta_{0s} + \beta_{1t} + \sum_{k=1966}^{1984} \gamma_k (G_s * T_k) + \delta \mathbf{X}_{st} + \epsilon_{st} \quad (1)$$

where HYV_{st} is the proportion of cultivated land with HYV seeds in state s at time t, G_s captures states' time-invariant measure of groundwater, and T_k is a year dummy equal to 1 whenever $k = t$.

To account for the availability of abundant groundwater that could be obtained with electric pumpsets and used to provide timely and adequate irrigation, I use the measure of groundwater described above. State controls, X_{st}, are included to control for characteristics suspected to be correlated with the diffusion of HYV seeds, such as the proportion of rural population, the log of real per capita education expenditure, the log of real per capita credit availability and

[5] Other measures of electrification include the connected load, i.e. maximum consumption available per consumer, measured in KW per million people; the number of industrial users connected to the network per 1000 people and the proportion of villages electrified.

[6] Sources and variables are all publicly available online from "EOPP Indian States Data", EOPP-STICERD, London School of Economics.

Table 2
States: groundwater, electrification and manufacturing output.

State	Groundwater[a]	Agricultural connections (per 1000 people)						Real manufacturing output (per capita)					
		Level	Change					Level	Change (%)				
	1965	1965	1965–70	1970–75	1975–80	1980–84	**1965–85**	1965	1965–70	1970–75	1975–80	1980–84	**1965–84**
Andhra Pradesh	**0.07**	1.48	2.98	1.77	2.20	2.25	**9.21**	675	16.57	18.56	17.87	14.02	**85.75**
Assam	**0.15**	0.00	0.00	0.05	0.05	0.04	**0.14**	1095	−41.26	29.75	−36.18	113.43	**3.81**
Bihar	**0.15**	0.21	0.98	0.74	0.39	0.31	**2.43**	610	−9.58	44.66	−17.15	82.44	**97.72**
Gujarat	**0.20**	0.74	1.95	1.57	2.71	1.15	**7.39**	1727	5.46	4.84	33.14	18.47	**74.41**
Haryana	**0.73**	1.56	7.34	3.95	5.13	1.47	**17.89**	1142	6.57	18.55	42.68	30.89	**135.95**
Karnataka	**0.00**	1.66	2.93	2.44	1.64	2.78	**9.79**	812	79.59	−4.12	1.56	14.98	**101.08**
Kerala	**0.00**	0.38	0.53	0.91	1.86	1.07	**4.37**	738	25.80	−0.09	50.75	−9.85	**70.82**
Madhya Pradesh	**0.00**	0.22	1.22	2.45	3.08	1.94	**8.68**	552	21.36	4.85	60.48	7.05	**118.59**
Maharashtra	**0.01**	1.10	3.30	2.82	3.69	3.08	**12.89**	2304	19.15	−7.32	57.81	−12.34	**52.76**
Orissa	**0.05**	0.01	0.01	0.13	0.46	0.32	**0.92**	718	−14.23	−19.63	103.34	4.40	**46.35**
Punjab	**0.80**	1.86	5.04	3.08	7.01	6.25	**21.38**	1063	13.08	42.99	5.56	29.60	**121.21**
Rajasthan	**0.01**	0.30	1.13	1.83	3.00	1.16	**7.12**	600	14.68	−8.19	19.68	21.96	**53.68**
Tamil Nadu	**0.06**	6.93	5.03	4.61	1.88	2.18	**13.70**	1559	14.07	−12.73	78.03	11.66	**97.87**
Uttar Pradesh	**0.45**	0.18	1.19	1.05	0.68	0.45	**3.36**	558	33.70	1.17	12.65	28.60	**95.95**
West Bengal	**0.35**	0.02	0.01	0.19	0.22	0.03	**0.46**	2037	−8.70	12.11	28.76	−4.78	**25.49**

[a] Groundwater is defined as the proportion of state area covered with aquifers thicker than 150 m.

population density. Additionally, state and time fixed effects are included and errors are robust to heteroskedasticity and autocorrelation. The coefficients of interest in Eq. (1) are γ_k, where positive and significant values would indicate that states with greater access to groundwater adopted a greater proportion of HYV seeds. Increasing values of γ_k would also indicate that the difference was growing over time, i.e. the appropriate characteristics for successful adoption would trigger a path of divergence across states in HYV adoption. The estimation of Eq. (1) provides point estimates of the yearly impact of groundwater availability on HYV adoption, as captured by the coefficients γ_k. Results in Fig. 3[7] show that states with greater groundwater availability have become relatively more intensive in the use of HYV seeds.

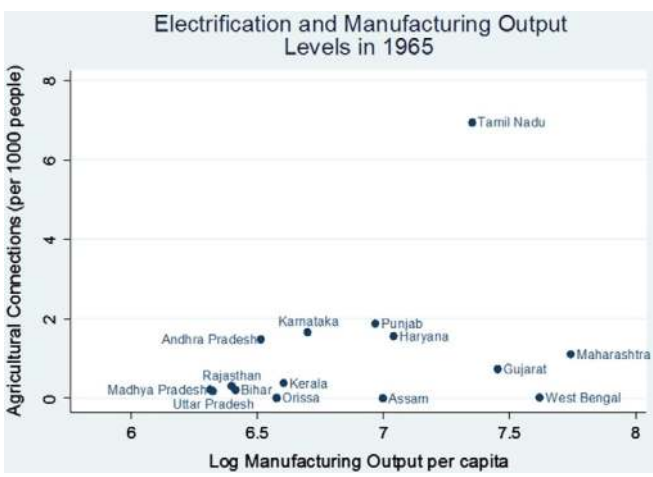

Fig. 1. Agricultural connections and manufacturing output: initial levels.

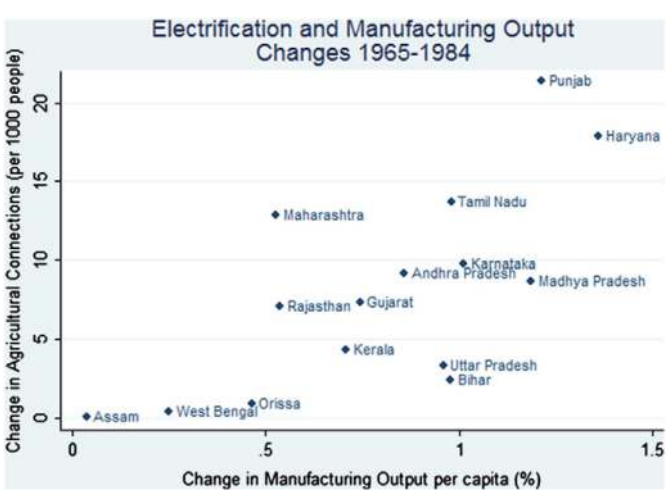

Fig. 2. Agricultural connections and manufacturing output: changes.

3.1.2. Groundwater, electrification and industrial output

The consistency of the IV estimator relative to OLS requires a strong correlation between the instrument (namely, G_s), and the instrumented variable, electricity provision (e_{st}).[8] To test this, I check whether the timing at which some states have improved their electricity reach significantly coincides with the start of the Green Revolution. Additionally, I perform a similar exercise to see whether a similar pattern emerges for the measure of industrialization used in the second stage of the IV strategy. As in the previous section, I use a specification where I estimate the time-varying effect of groundwater availability on outcomes (Y_{st}), whether electrification or manufacturing output:

$$Y_{st} = \beta_{0s} + \beta_{1t} + \sum_{k=t}^{1984} \gamma_k (G_s * T_k) + \delta \mathbf{X}_{st} + \epsilon_{st} \tag{2}$$

Figs. 4 and 5 show, respectively, the results for electrification and for manufacturing output using data available from 1960 only.[9] Results show that states with more groundwater have expanded the electricity reach in rural areas, significantly only after the Green Revolution. The specification in Eq. (2) imposes no constraint on the functional form of the time-varying effects of groundwater on electrification. However, the coefficients in Fig. 4 suggest that from 1965 onwards a linear specification would be an adequate fit to capture the divergence in electrification. In terms of the reduced form specification, Fig. 5 also suggests a slight divergence in manufacturing output after the start of the Green Revolution, in states with more abundant groundwater.

[7] Results are not sensitive to the exclusion of state controls.
[8] In addition, IV requires the exclusion restriction to hold. I explore this below.
[9] Results show a similar pattern for specifications without state controls.

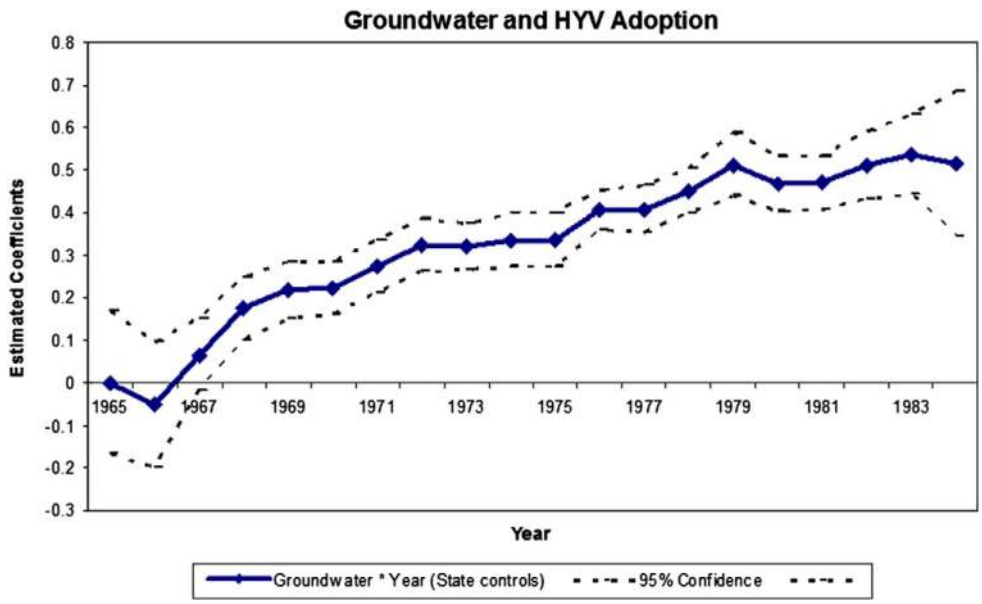

Fig. 3. Groundwater and HYV adoption. These results report coefficients obtained from a regression of HYV Adoption at the state level on the interaction between the measure of groundwater availability and year dummies. The regression controls for state and year fixed effects and for state controls (log of education expenditure per capita, share of rural population, population density and log of total credit per capita). "Groundwater" is the proportion of state area covered with aquifers thicker than 150° m. "HYV Adoption" is the proportion of cropped area with HYV seeds. Standard errors are robust to heteroskedasticity.

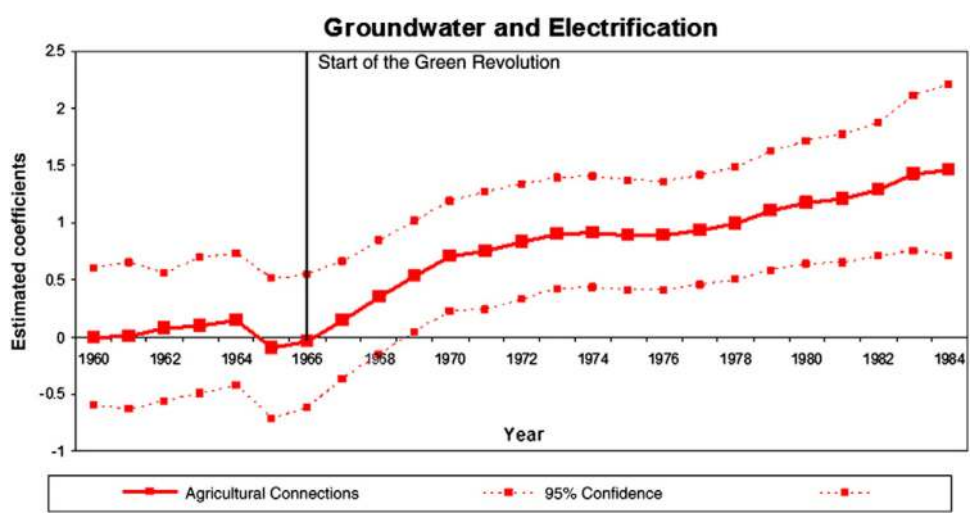

Fig. 4. Groundwater and electrification. These results report coefficients obtained from a regression of Agricultural Connections at the state level on the interaction between the measure of groundwater availability and year dummies. The regression controls for state and year fixed effects and for state controls (log of education expenditure per capita, share of rural population, population density and log of total credit per capita). "Agricultural Connections" is the number of agricultural units connected to the electricity network (per 1000 people). "Groundwater" is the proportion of state area covered with aquifers thicker than 150° m. Standard errors are robust to heteroskedasticity.

3.1.3. Evidence at the district level

In this section, I check the robustness of the first stage using information on the total number of pumpsets per capita available at the district level. In particular, I run a similar set of regressions as previously where the divergence in pumpset adoption (P_{dst}) is explained by the interaction of groundwater availability and year dummies. Even though there is no information on availability of electricity at the district level, the anecdotal evidence suggests a shift from diesel to electrified pumpsets after the start of the Green Revolution (see for example, Bharadway, 1990 or McGuirk and Mundlak, 1991). To add confidence to the use of the Green Revolution as a natural experiment, I perform a similar regression where I replace the explained variable by a measure of HYV adoption.

$$P_{dst} = \beta_{0ds} + \beta_{1t} + \sum_{k=1966}^{1984} \gamma_k (G_{ds} * T_k) + \delta_t \mathbf{X}_{dst} + \epsilon_{dst} \quad (3)$$

As in the analysis for the state level, G_{ds} is a measure of groundwater availability, in this case a dummy that equals 1 if the district has aquifers thicker than 150 m and 0 otherwise. The estimated γ_k provides the average conditional difference on pumpset use in year k between districts with an aquifer thicker than 150 mts and those with shallower aquifers. I exploit within district variation by using district fixed effects. I also include year dummies and run Eq. (3) with district controls X_{dst}.[10]

Fig. 6 plots the estimated γ_k for two specifications. As shown for the state level regressions, districts with greater groundwater abundance are associated with an increasing use of irrigation pumpsets

[10] Time varying controls at the district level include literacy, population density, and proportion of rural population. Standard errors are clustered at the district level.

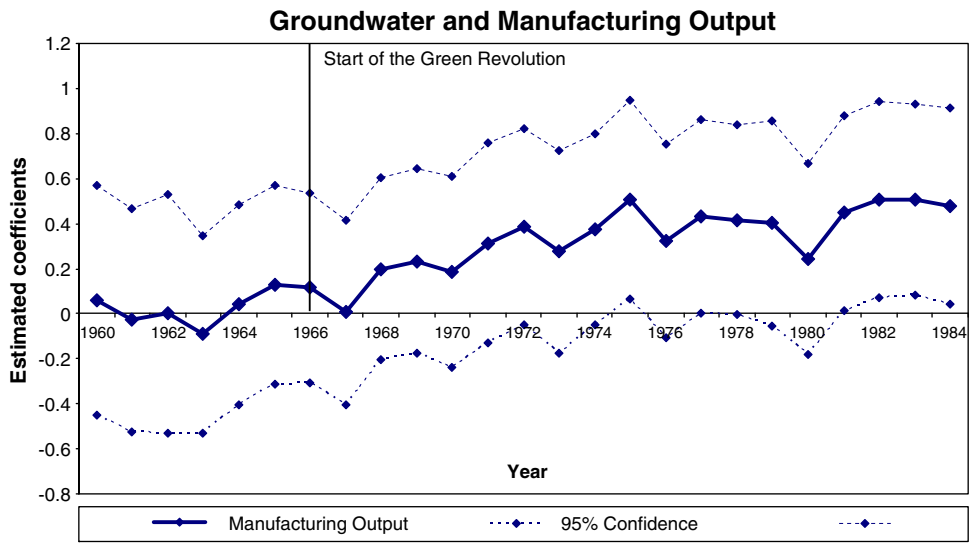

Fig. 5. Groundwater and manufacturing output. These results report coefficients obtained from a regression of log real manufacturing output per capita at the state level on the interaction between the measure of groundwater availability and year dummies. The regression controls for state and year fixed effects and for state controls (log of education expenditure per capita, share of rural population, population density and log of total credit per capita). "Groundwater" is the proportion of state area covered with aquifers thicker than 150° m. Standard errors are robust to heteroskedasticity.

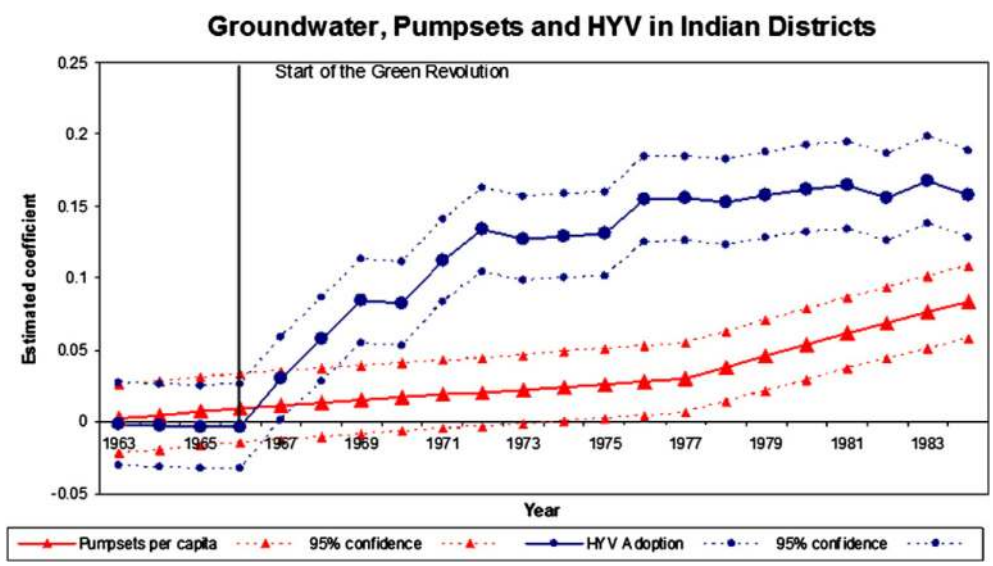

Fig. 6. Differential pumpset use and HYV adoption in districts with deep groundwater. These results report coefficients obtained from regressions of pumpsets per capita and HYV adoption at the district level on the interaction between the measure of groundwater availability at the district level and year dummies. The regressions control for district and year fixed effects and for district controls (share of literate population, share of rural population and population density). "Groundwater" is a dummy equal to 1 if the district has aquifers thicker than 150 mts. "HYV adoption" is the proportion of cropped area with HYV seeds. "Pumpsets per capita" is the number of powered pumpsets per district population.

and HYV seeds.[11] Estimates for the interaction terms are positive, significantly different from 0 only after the Green Revolution and increasing, meaning that there is a positive marginal effect for districts with more groundwater. This set of results is consistent with the findings at the state level and provides more evidence validating the choice of instruments.

In Table 3 I check the link between deep aquifers and a measure of rural power availability, available at the district level, that was collected by the Ministry of Agriculture and Irrigation of India (1976) only for 1974. Columns (1) and (2) confirm the link between groundwater and rural power. To illustrate this point further I look at the states of Uttar Pradesh and Bihar, since they provide a good testing ground of the link between groundwater and use of pumpsets and power availability. I can compare districts within two states in a region of India where deep aquifers are adjacent to areas with no groundwater (categories 1 and 4, respectively, according to the National Atlas of India classification discussed above). When looking at the measure of power availability and operating pumpsets in Columns (3) to (6) in Table 3, the districts with more groundwater availability show significantly higher measures of rural power. Overall, the district evidence is in line with the idea that access to deeper aquifers facilitated the expansion of the electricity network.

3.2. OLS and IV results

In this section I explore further the whether the reduced form relationship between groundwater and industrial outcomes can be robustly ascribed to the expansion of the electricity network across

[11] Results hold in specifications without controls as well.

Table 3
Groundwater and power availability at the district level.

	(1)	(2)	(3)	(4)	(5)	(6)
	Power availability (1974)		Power availability (1974)		Total pumpsets (1981)	
	All India		Uttar Pradesh and Bihar			
Groundwater	**0.142**	**0.174**	**0.068**	**0.104**	**0.006**	**0.011**
	(0.034)***	(0.47)***	(0.032)**	(0.042)**	(0.003)**	(0.004)**
Literacy		**0.297**		**0.250**		**0.012**
		(0.13)**		(0.19)		(0.02)
Rural population		**−0.387**		**−0.256**		**−0.047**
		(0.11)***		(0.14)**		(0.01)***
Population density		**−0.002**		**0.005**		**−0.007**
		(0.01)*		(0.01)		(0.001)
Observations	254	254	65	65	65	65

Robust standard errors in parentheses. "Power availability" is a measure, in horses per hectare, of power in rural areas. "Total pumpsets" is a measure of reported pumpsets per capita. "Groundwater" is a dummy equal to 1 if the district has aquifers thicker than 150 mts. "Population density" is total population per squared kilometer. "Literacy" and "rural population" are measured as a proportion of total population.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

Indian states. To that end, I start by estimating an OLS regression where manufacturing output (y_{st}) is expressed as a function of the measure of electricity supply (e_{st}):

$$y_{st} = \alpha_{0s} + \alpha_{1t} + \lambda e_{st} + \alpha_3 \mathbf{X}_{st} + \mu_{st} \tag{4}$$

The regression includes time varying state controls (X_{st}), state fixed effects included to control for persistent and constant features within states, and year fixed effects to control for shocks common to all states. In particular, in the most parsimonious specification I control for demographic features (i.e. log of population density, proportion of rural population), credit availability (log of real per capita total credit) and a measure of development expenditure at the state level (log of real per capita expenditure in education). I discuss below a fuller specification.

The use of state fixed effects allows me to control for all state characteristics that remain constant over time, including unobserved variables that would contaminate the standard errors. Results discussed throughout are robust to the presence of heteroskedasticity and to different degrees of autocorrelation.[12]

In the OLS estimation of the coefficient of interest, λ, endogeneity is a major concern for quantifying the effect of infrastructure on output. The presence of correlation between the error term and the explanatory variable could be explained by reverse causality (e.g. states that are industrializing faster demand investments in expanding the electricity network) or omitted variables (e.g. unobserved changes in the institutional environment that drive both industrialization and electrification), and would introduce a bias to the estimation. However, I am agnostic about the sign of the bias, since alternative stories could work in both directions. For example, phenomena such as a pro-business environment or other political economy considerations could explain a positive correlation with both the outcome and the explanatory variable, biasing the estimate upwards. Alternatively, sources of bias towards zero could be any unobserved process driving rural electrification positively, but at the expense of industrial producers (such as skewed electricity pricing policies or rural lobbies) or the presence of attenuation bias because of measurement error in the electricity indicator (produced by illegal connections, theft of electricity, etc. that tend to be widespread in rural areas, see Tongia, 2003).

For that reason I subsequently run the IV specifications where e_{st} is replaced by $\hat{e}_{st} = f(G_s * Year_t; \mathbf{X_{st}})$. That is, I instrument the electricity indicator using the measure of groundwater availability that has shown explanatory power with respect to states' divergence in both HYV diffusion and electricity expansion. Note that the specification in the previous section does not impose a functional form on the divergence in electrification for states with more groundwater available. However, Fig. 4 suggests that the divergence process might have followed a linear trend. In fact, results using different specifications show that the interaction of G_s with a linear time-trend provides the strongest first stage (see Table 9 for quadratic and fully flexible specifications). For that reason, in what follows I present results using the linear first stage.

Results are shown in Table 4. Columns (1) and (2) show the OLS regressions with and without state controls, respectively, and find a positive and significant correlation between electrification and industrialization. Since the electrification process was targeted at agricultural producers, it is reassuring to find a positive and strongly significant correlation between electrification and industrial output once other time-varying factors and state fixed effects are controlled for. Similarly, columns (3) and (4) show results of the reduced form regressions of the instrument on manufacturing output. The interaction between groundwater and a time trend is, in both cases, positive and significant at the 1% level.

Columns (5) to (7) show the key results in this paper, where I use the time varying effects of groundwater availability as an instrument for electrification. Columns (5) and (6) show results without and with state controls, respectively. Column (7) shows results with controls when accounting for first degree autocorrelation.[13] In all cases, the coefficients are positive and strongly significant. Also, coefficients are slightly greater than OLS estimates, suggesting the net presence of a downward bias. At the bottom of Table 4 details of the first stage are presented. The coefficient on the interaction is positive and significant in all cases and the F-statistics show large values that confirm the strength of the instrument.

The IV results could be interpreted as a local average treatment effect: states that increased their rural electrification in 1 connection per 1000 people because of their groundwater endowment would experience around a 0.027 log points average increase in their manufacturing output per capita. In particular, there could be other margins of electrification that are not captured by this instrument, e.g. if a state with no groundwater has electrified for other reasons. For example, Maharashtra or Rajasthan show substantial increases in their measure of electrification but almost no groundwater. The variation in this exercise only comes from non-zero groundwater states. This constitutes an important caveat to the interpretation and generalization of these results.

[12] As pointed out by Wooldridge (2003) and Bertrand et al. (2004), clustering standard errors would not be appropriate to deal with within-state autocorrelation when the number of states is small. To address this issue in a more general way I also perform successfully a wild cluster bootstrap-t estimation, as in Cameron et al. (2008), that has proven to work well with small number of clusters. Results are available upon request.

[13] Using Newey–West standard errors, where the error structure is assumed to be heteroskedastic and arbitrarily autocorrelated.

Table 4
Main results: OLS and IV.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Log manufacturing output						
	OLS		Reduced form		IV		
Agricultural connections	**0.029**	**0.022**			**0.038**	**0.027**	**0.027**
	(0.005)***	(0.005)***			(0.008)***	(0.007)***	(0.008)***
Groundwater × time trend			**0.022**	**0.020**			
			(0.004)***	(0.005)***			
Education expenditure		**0.155**		**0.144**		**0.167**	**0.167**
		(0.060)**		(0.062)**		(0.065)**	(0.074)**
Rural population		**−7.763**		**−12.75**		**−7.001**	**−7.001**
		(3.251)**		(3.525)***		(3.787)**	(4.702)
Population density		**1.223**		**0.761**		**1.172**	**1.172**
		(0.600)*		(0.785)		(0.647)*	(0.773)
Total credit		**0.026**		**0.015**		**0.024**	**0.024**
		(0.048)		(0.051)		(0.045)	(0.051)
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	300	300	300	300	300	300	300
First stage							
					Agricultural connections		
Groundwater × time trend					**0.591**	**0.735**	**0.735**
					(0.101)***	(0.082)***	(0.107)***
Education expenditure						**−0.841**	**−0.841**
						(0.993)	(1.175)
Rural population						**−214.1**	**−214.1**
						(19.78)***	(25.47)***
Population density						**−15.28**	**−15.28**
						(5.943)**	(7.761)**
Total credit						**−0.356**	**−0.356**
						(0.430)	(0.555)
First stage F-test					34.20	79.01	46.90
R-square					0.95	0.95	0.95

Standard errors in parentheses, robust to heteroskedasticity. "Agricultural connections" is the number of agricultural units connected to the electricity network per 1000 people. "Log manufacturing output" is the log of real per capita manufacturing output. "Groundwater" is the proportion of the state area covered with aquifers thicker than 150 mts. "Education expenditure" and "total credit" are in logs, per capita. "Rural population" is a share total population and, "population density" is population (in 1000 s) per square kilometer. F-test: Staiger and Stock's (1997) rule of thumb is that instruments are "weak" if the first-stage F is less than 10, the Stock and Yogo (2003) Weak ID test critical value for 2SLS bias being less than 10% of OLS bias is 16.38. Column (7) shows Newey–West standard errors assuming first-degree autocorrelation. Results are robust to higher degrees of autocorrelation. All data are for 1965–1984.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

The size of the estimates of λ also suggest that the impact of electrification is substantial: an increase in one standard deviation in the number of rural connections implies an increase of around 14.7% in manufacturing output. It is important to note that, given the great variation in electricity provision across states, a state at the mean of the electrification distribution would need to double its rural electrification to observe such an effect. Another idea of the magnitude of the effect can be taken from asking what the benefit for a state at the 25th percentile of the distribution of electrification would be to move to the 75th percentile. For example, Bihar in 1984 had a real manufacturing output per capita of Rs 1206 and 2.64 farms connected to the network per 1000 people. If Bihar engaged in a rural electrification process to move to the top 25% states in terms of electrification, i.e. increasing the number of connections per 1000 people by 8.8,[14] the model predicts that the manufacturing output would increase by 23.7%, or around 24 billion Rupees.[15] This would constitute an increase of around 4.2% of Bihar's total gross product per year, that can be used as a benchmark for how much the state could spend in a rural electrification program.[16]

4. Possible alternative channels

The main concern with respect to the identification strategy at hand is related to the exclusion restriction. This means that if the measure of rural electrification is strongly correlated with other state outcomes that are actually driving the industrialization process, then the results obtained in the previous section would be spurious. The use of observational data makes the task of validating the exclusion restriction very hard, since even though the characteristics of the water table in a state could be as good as random, it is not possible to rule out all alternative channels that could explain the complex process of industrialization.

In this section I discuss the likely alternative channels that might be explaining the reduced form results and I contrast them to the electrification channel proposed in this paper. To do so, I perform three different checks. I first explore whether the electrification channel holds when exposed to variation in other observable state characteristics associated to mechanisms that the economic literature has underlined as important in the process of industrial growth. I subsequently take a step back in terms of the electrification channel and put to the data the following question: could the same strategy be used to explain divergence in industrial development through other means than electrification? In short, the second set of checks consists in running OLS and IV specifications with these alternative stories as in Table 4 to see whether the first and second stages hold across specifications. The third and final set of checks implies looking at the

[14] That is an increase of around 650,000 connections, up from 200,000.

[15] This is obtained by multiplying the estimated increase in real manufacturing output per capita times total population in Bihar in 1984 (around 74 million people). The value is in 1974 Rupees.

[16] Note that this measure does not take into account the effect of electrification in other economic activities, including agriculture.

composition of industrial outcomes, that could provide suggestive evidence against the electrification mechanism.

4.1. Alternative channels: a brief discussion

There are many stories in the literature as to why shocks to agricultural productivity would spill over to the manufacturing sector. Johnson (1997) points out that a rapid increase in agricultural productivity contributed to the start of the Industrial Revolution, by releasing (cheap) labor for the manufacturing sector and creating the agricultural surplus needed to feed the growing population. This argument could be linked to a process of urbanization as the agricultural sector lays workers off. To check this I use the proportion of urban population in a state. Matsuyama (1992) formalizes these mechanisms in a model where both sectors compete for labor and where exogenous increases in agricultural productivity benefit the manufacturing sector. This mechanism works only in closed economies, since comparative advantages in the agricultural sector might deprive industries from labor in open economies. In that case, effects on industrialization could go either way.[17] Murphy et al. (1989a) propose a mechanism wherein industrialization happens because positive shocks to agricultural productivity boost rural income and – subsequently – the demand for manufactures. This result hinges on two conditions: non-homothetic preferences (the share of expenditure on manufactures increases with income) and a fair distribution of benefits from increases in agricultural productivity to sustain a sizeable demand.[18] Information on the mean expenditure of rural households allows me to control for this channel.

Other stories might imply the development of complementary markets, such as credit markets. For example, the increase in rural incomes affects saving rates in rural areas and, subsequently, credit availability. To check this, I use information on non-agricultural rural credit. Similarly, Foster and Rosenzweig (1996) have shown greater private investments in schooling after the Green Revolution has increased returns to education. I use the proportion of literate population, that might also be at the heart of improvements in the manufacturing sector.

Finally, there is a political economy channel that raises two issues: the first one is why farmers' demand for electricity was satisfied in groundwater (and HYV) intensive states. Second, whether these political circumstances that facilitated electrification can also explain, independently, improvements in the manufacturing sector.

Two mechanisms could explain why farmers' need of electricity was satisfied by their state governments. The first one is linked to the preferences of the median voter. In the period analyzed, rural population in any given state is at least 65% of total population, and almost 80% on average (reaching 90% in three states), in a country that has been democratic and federal since its independence in 1947. There is some evidence that voters in states intensive in HYV like Punjab, Haryana and Tamil Nadu moved away from the Congress Party and voted for regional parties. A second mechanism is supported by anecdotal evidence: the endogenous formation of lobbies. It seems important to note that "increasing food shortages and mounting concern for immediate gain in production led to the shift in developmental priorities" (Sharma and Dak, 1989), such that expanding the production possibilities in the agricultural sector was deemed fundamental. That circumstance gave farmers, in particular big ones, an unprecedented political clout over state governments. Dhanagare (1989) estimates that "the prosperity unleashed by the Green Revolution was distributed differentially, putting the small and marginal farmers at a relative disadvantage. The high cost/high yield technology called for capital investments beyond the means of a majority of small and marginal farmers." It might be the case that a stronger economic position translated into political leverage. Tongia (2003), for example, stresses that rural electrification "has swayed in strong political winds."Gulati and Narayanan (2003) also show that what they call 'the subsidy syndrome in Indian agriculture' after the introduction of the new seeds in late 60s became a fundamental instrument of economic policy where available electricity was one of its main channels.

With respect to direct effects on manufacturing outcomes, there are many channels that could operate. For example, as noted by Laffont (2005), government inefficiencies and corruption increase the marginal cost of raising funds and constrain the ability of the executive to invest in infrastructure. Alternatively, other phenomena such as the power of industrial lobbies or unions, or the social roots of a party might explain why resources are shifted in one or another way. It follows that the indicators of electricity network development might actually be capturing these political economy features that could also drive the dependent variable.

To address the concern that the political economy channel is driving the results, I include state controls that capture political party outcomes (party allegiance of the Chief Minister and share of the votes to the main national party, the Congress Party of India). Additionally, I also use expenditure in development sectors (infrastructure, health and education) that can capture not just states' investment in physical and human capital, but also the ability and/or willingness of the state government to enhance the economic environment. Similarly, I use the labor regulation measured constructed by Besley and Burgess (2004) to capture whether the states have enacted more pro-worker or pro-business regulation over time.

4.2. Alternative channels as additional controls

In Table 5 I progressively include the above-described variables as controls in both stages of the IV specification and check whether the electrification channel holds.

In all specifications in columns (1) to (5), the coefficient on electrification is not only significant but also its magnitude remains quite stable, albeit lower than in Table 4. The inclusion of these variables does not reduce significantly the explanatory power of the electricity story.

The coefficients on the control variables do not provide much additional information. In fact, the political economy story seems to be the one providing more explanatory power to differences in manufacturing output. First, results suggest that an active state, as captured by the development expenditure for example, is always positive and significant. Column (5) also shows that, as in Besley and Burgess (2004), more pro-worker regulation is negatively correlated with manufacturing outcomes. These two results suggest that lobbying might indeed have had an impact in industrial outcomes. A similar picture emerges when looking at electoral outcomes. States where the Congress Party gets more votes and run the state government or states that vote for a leftist government seem to industrialize more than states that vote for the Janata or other parties. Note that the use of state fixed effects implies that these results do not come only from the cross-section of states.

Column (6) controls for the time-varying effects of pre-Green Revolution levels of manufacturing output, literacy and infrastructure (i.e. roads) and finds a positive and significant coefficient on the measure of electrification. This suggests that the results are not driven by initial differences found in states with deep aquifers that might have explained the divergence in electrification or HYV adoption.

Overall, the coefficient on electrification remains significant and similar in magnitude, suggesting the electrification channel remains strong. It has to be noted, however, that this evidence does not preclude the possibility that alternative unobserved time-varying phenomena with enough idiosyncratic variation could be at play. In that sense, these results only suggest that, with the information at hand, the electrification channel remains robust.

[17] India, being a very large federal country, could be thought as a collection of open economies. In that case, the mechanisms described by Matsuyama might be present.

[18] See also Voigtlander and Voth (2006). This could undermine the argument when applied to India, since it is well documented (see for example, Dhanagare, 1989) that the Green Revolution has increased the rift between poor and rich farmers and, in some cases, has not even increased poor peasants' real incomes.

Table 5
Main results with additional controls.

	(1)	(2)	(3)	(4)	(5)	(6)
	Log manufacturing output					
	IV using groundwater × time trend					
Agricultural connections	**0.020**	**0.020**	**0.021**	**0.020**	**0.024**	**0.034**
	(0.007)***	(0.007)***	(0.008)***	(0.007)***	(0.008)***	(0.010)***
Development expenditure	**0.331**	**0.330**	**0.329**	**0.335**	**0.309**	**0.313**
	(0.073)***	(0.073)***	(0.071)***	(0.072)***	(0.071)***	(0.071)***
Literacy		**−0.325**	**−0.365**	**−0.510**	**−0.582**	**−1.398**
		(1.121)	(1.121)	(1.289)	(1.073)	(1.296)
Mean HH rural expenditure			**0.033**	**0.055**	**0.030**	**−0.041**
			(0.117)	(0.118)	(0.113)	(0.121)
Non agricultural rural credit				**0.283**	**0.242**	**0.324**
				(0.123)**	(0.124)*	(0.127)***
Labor regulation					**−0.047**	**0.016**
					(0.019)**	(0.022)
Vote share Congress party					**0.004**	**0.004**
					(0.001)***	(0.001)***
Chief Minister Congress party					**0.076**	**0.071**
					(0.031)**	(0.032)**
Chief Minister hard left party					**0.125**	**0.097**
					(0.055)**	(0.058)*
Chief Minister Janata party					**0.041**	**0.060**
					(0.034)	(0.044)
Initial manufacturing income time-trend						**37.21**
						(9.047)***
Initial literacy time-trend						**0.077**
						(0.026)***
Initial roads time-trend						**−0.004**
						(0.016)
Education expenditure	**−0.102**	**−0.105**	**−0.102**	**−0.102**	**−0.104**	**−0.090**
	(0.080)	(0.081)	(0.082)	(0.078)	(0.076)	(0.077)
Rural population	**−4.126**	**−4.515**	**−4.329**	**−4.458**	**−4.383**	**1.540**
	(3.502)	(3.411)	(3.285)	(3.223)	(3.050)	(3.291)
Population density	**0.701**	**0.845**	**0.813**	**1.227**	**1.778**	**1.197**
	(0.587)	(0.738)	(0.724)	(0.730)*	(0.719)**	(0.697)*
Total credit	**0.033**	**0.035**	**0.036**	**−0.218**	**−0.211**	**−0.421**
	(0.043)	(0.043)	(0.043)	**(0.117)***	(0.118)*	(0.141)***
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	300	300	300	300	300	300
First stage						
	Agricultural connections					
Groundwater × time trend	**0.703**	**0.665**	**0.707**	**0.715**	**0.709**	**0.652**
	(0.088)***	(0.083)***	(0.080)***	(0.078)***	(0.078)***	(0.089)***
First stage F-test	62.60	64.87	77.08	83.05	41.41	87.70
R-square	0.95	0.95	0.95	0.95	0.96	0.98

Standard errors in parentheses, robust to heteroskedasticity. "Agricultural Connections" is the number of agricultural units connected to the electricity network per 1000 people. "Log manufacturing output" is the log of real per capita manufacturing output. "Groundwater" is the proportion of state area covered with aquifers thicker than 150 mts. "Education expenditure", "total credit", "development expenditure", and "non agricultural rural credit" are in logs, per capita. "Rural population" and "literacy" are shares of total population and "population density" is population (in 1000 s) per square kilometer. "Labor Regulation" is taken from Besley and Burgess (2004). All "Chief Minister" variables are dummies equal to 1 if the chief minister belongs to the parties mentioned. All "initial" variables are measured in 1965. "Roads" is surfaced state roads (in kms) indexed by population density. F-test: Staiger and Stock's (1997) rule of thumb is that instruments are "weak" if the first-stage F is less than 10, the Stock and Yogo (2003) Weak ID test critical value for 2SLS bias being less than 10% of OLS bias is 16.38. Coefficients on all covariates in the first stage are available from the author. All data are for 1965–1984.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

4.3. Groundwater as an instrument for alternative channels

In this section I investigate further if some of the alternative stories could have been explaining the reduced form results documented in the previous section. Basically, whether the time-varying effect of groundwater availability is a strong predictor of variation in these variables in the first stage and, if so, if the predicted variation can explain manufacturing outcomes in the second stage. Even though variation in groundwater was presented as very specific to the electrification channel, because of the use of electrified pumpsets, it might well be the case that it is affecting manufacturing output through any of the channels associated to the HYV agricultural productivity shock.

Table 6 presents OLS and IV results. I am interested in testing whether the estimates show the expected sign and similar levels of significance for the point estimates and the first stage F-tests.

The main conclusion is that none of the alternative stories passes all tests consistently. In some cases, e.g. literacy, rural expenditure or rural credit, the OLS estimates are not significant. However, it could well be

Table 6
Alternative explanations.

Panel A: OLS	(1)	(2)	(3)	(4)	(5)	(6)
	Log manufacturing output					
Agricultural connections	**0.022**					**0.019**
	(0.005)***					(0.005)***
Development expenditure		**0.402**				**0.341**
		(0.07)***				(0.07)***
Literacy			**1.197**			**−0.376**
			(1.07)			(0.12)
Mean HH rural expenditure				**0.064**		**0.056**
				(0.14)		(0.13)
Non agricultural rural Credit					**0.120**	**0.269**
					(0.17)	(0.14)**
Rural population	**−7.763**	**−8.049**	**−11.87**	**−13.29**	**−13.58**	**−4.65**
	(3.25)**	(3.21)**	(3.88)***	(3.37)***	(3.52)***	(3.34)
Population density	**1.223**	**0.794**	**0.913**	**1.362**	**1.617**	**1.155**
	(0.68)*	(0.63)	(0.87)	(0.71)*	(0.67)**	(0.78)
Education expenditure	**0.165**	**−0.155**	**0.185**	**0.180**	**0.176**	**−0.105**
	(0.06)***	(0.08)*	(0.07)***	(0.07)***	(0.07)***	(0.08)
Total credit	**0.026**	**0.042**	**0.026**	**0.036**	**−0.073**	**−0.205**
	(0.05)	(0.05)	(0.05)	(0.05)	(0.15)	(0.12)*
R-square	0.94	0.95	0.94	0.94	0.94	0.95
Observations	300	300	300	300	300	300

Panel B: IV using groundwater × time trend	(1)	(2)	(3)	(4)	(5)	(6)
	Log manufacturing output					
Instrumented variable						
Agricultural connections	**0.027**					
	(0.007)***					
Development expenditure		**2.238**				
		(5.67)				
Literacy			**−6.61**			
			(7.89)			
Mean HH rural expenditure				**−2.307**		
				(1.15)**		
Non agricultural rural credit					**11.52**	
					(24.5)	
Rural population						**−73.39**
						(37.8)*
Control variables						
Rural population	−7.701	12.47	−14.84	−24.12	−13.86	
	(3.79)*	(54.2)	(7.97)*	(6.31)***	(12.85)	
Population density	**1.172**	**−2.015**	**4.088**	**4.741**	**17.09**	**9.049**
	(0.65)*	(8.30)	(3.26)	(1.70)***	(32.7)	(4.82)*
Education expenditure	**0.167**	**−1.653**	**0.419**	**−0.010**	**0.341**	**0.525**
	(0.07)**	(4.67)	(0.28)	(0.15)	(0.55)	(0.26)**
Total credit	**0.024**	**0.058**	**−0.170**	**−0.003**	**−10.35**	**−0.079**
	(0.45)	(0.06)	(0.13)	(0.08)	(22.1)	(0.12)
Observations	300	300	300	300	300	300

Panel C: first stage	Agricultural connections	Development expenditure	Literacy	Mean HH rural expenditure	Non agricultural rural credit	Rural population
Groundwater × time trend	**0.735**	**0.003**	**−0.0008**	**−0.009**	**0.002**	**−0.0003**
	(0.08)***	(0.01)	(0.0005)*	(0.003)***	(0.004)	(0.0001)*
Rural population	**−214.1**	**−9.445**	**−0.932**	**−4.926**	**0.096**	
	(19.8)***	(2.30)***	(0.20)***	(0.94)***	(1.20)	
Population density	**−15.28**	**1.369**	**0.459**	**1.725**	**−1.418**	**0.137**
	(5.94)**	(0.60)**	(0.05)***	(0.31)***	(0.48)***	(0.02)***
Education expenditure	**−0.841**	**0.809**	**−0.008**	**−0.067**	**−0.017**	**0.006**
	(0.99)	(0.07)***	(0.01)	(0.03)**	(0.04)	(0.002)***
Total Credit	**−0.356**	**−0.027**	**0.008**	**−0.008**	**0.900**	**−0.002**
	(0.43)	(0.04)	(0.003)***	(0.02)	(0.03)***	(0.002)
First stage F-test	79.01	0.13	3.42	8.07	0.19	3.79
R-square	0.95	0.97	0.99	0.89	0.99	0.99

Table 7
Industry composition.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Log manufacturing output							
	IV using groundwater × time trend							
	Registered sector	Unregistered sector	High electricity	Low electricity	Food sector	No food	High water use	Low water use
Agricultural connections	**0.052**	**0.001**	**0.212**	**0.125**	**0.093**	**0.141**	**0.088**	**0.136**
	(0.015)***	(0.014)	(0.069)***	(0.037)***	(0.031)***	(0.046)***	(0.048)*	(0.038)***
Education expenditure	**0.387**	**0.084**	**−0.013**	**−0.183**	**0.052**	**0.141**	**−0.229**	**0.030**
	(0.220)*	(0.141)	(0.281)	(0.136)	(0.142)	(0.178)	(0.182)	(0.168)
Rural population	**−0.905**	**−12.80**	**46.68**	**16.70**	**12.37**	**16.81**	**5.429**	**16.45**
	(3.806)	(6.310)**	(16.70)****	(8.670)*	(7.521)*	(11.10)	(12.11)	(8.621)*
Population density	**−1.880**	**5.120**	**−7.417**	**−3.436**	**−3.011**	**−2.624**	**−2.650**	**−3.441**
	(0.770)**	(1.026)***	(3.176)**	(1.941)*	(2.100)	(2.285)	(3.010)	(1.846)*
Total credit	**0.158**	**−0.180**	**−0.099**	**0.159**	**0.308**	**0.183**	**−0.030**	**0.140**
	(0.062)***	(0.075)**	(0.266)	(0.151)	(0.147)**	(0.179)	(0.222)	(0.161)
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	300	300	180	180	180	180	180	180
First stage F-test	79.01	79.01	22.52	22.52	22.52	22.52	22.52	22.52

Standard errors in parentheses, robust to heteroskedasticity. Results are robust to different degrees of autocorrelation. All regressions include state and year fixed effects. "Agricultural connections" is the number of agricultural units connected to the electricity network per 1000 people. "Groundwater" is the proportion of state area covered with aquifers thicker than 150 mts. "Registered sector" includes factories with more than 10 workers and power or more than 20 workers without power. "Electricity use" and "water use" are high if the input use is above median for all Indian industries and low otherwise. "Education expenditure" and "total credit" are in logs, per capita. "Rural population" is a share total population and, "population density" is population (in 1000 s) per square kilometer. For columns (3)–(8) the coefficient of "groundwater * time trend" in the first stage is 0.745, with a standard error of 0.157. Full first stage available from the author. Stock and Yogo (2003) Weak ID test critical value for 2SLS bias being less than 10% of OLS bias is 16.38. Data is 1965–1984 for columns (1) and (2) and 1973–84 for all other columns.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

the case that the estimates are biased and the IV regressions would reverse that pattern. The three IV specifications in Panel B show that this is not the case, showing estimates that are not significantly different from 0. In terms of the first stage, as shown in Panel C, in some specifications (e.g. literacy, urbanization and rural expenditure) the interaction coefficient has a negative sign when a positive one was expected. Most importantly, in all cases the first stage would be weak: all F-statistics are below critical levels. The time-varying effects of groundwater availability provide a strong first stage for the measure of rural electrification, but it does not for any of the alternative channels.

These results increase the confidence that the variation used in the instrumental strategy works through the proposed channel and not through others. However, it does not mean that the alternative channels are unimportant or did not reflect differences across states, but simply that the variation used to instrument for the measure of electrification cannot be used to explain variation in urbanization rates, credit availability, literacy, state expenditure in development or expenditure in rural households. The use of state fixed effects in all specifications means that I exploit within state variation, and only in the case of the electricity measure the time-varying effect of groundwater availability has explanatory power for understanding states' divergence in industrial development.

4.4. Industry composition

In this section I break down manufacturing output in different ways, to test whether the effects on aggregate manufacturing outcomes can be associated with improvements in access to electrification or to other channels associated with the Green Revolution (such as an increase in food production) or to groundwater availability. In the first split of the data, I make use of the distinction in India between the formal (or 'registered') sector (i.e. firms with more than 10 workers that use power or more than 20 workers without power) and the informal (or 'unregistered') sector of smaller firms. Since the use of electricity is an important distinction between the two, I would expect the manufacturing output of the 'registered' sector to be more affected by improvements in access to electricity than the 'unregistered' sector. Columns (1) and (2) in Table 7 show that this is the case. The expansion of the electricity network is associated with increases in manufacturing output among firms in the formal

Notes to Table 6:
Standard errors in parentheses, robust to heteroskedasticity. Results are robust to different degrees of autocorrelation. All regressions include "State" and "Year" fixed effects, "agricultural connections" is the number of agricultural units connected to the electricity network per 1000 people. "Log manufacturing output" is the log of real per capita manufacturing output. "Groundwater" is the proportion of state area covered with aquifers thicker than 150 mts. "Mean household rural expenditure" is the log average expenditure in rural households, "Non agricultural rural credit" is a log measure of real per capita rural credit not used for agricultural projects and "development expenditure" is the log of real per capita expenditure by the state government in health, education and other development projects. "Literacy" and "rural population" are proportions of total population. "Education expenditure" and "total credit" are in logs, per capita. "Population density" is population (in 1000 s) per square kilometer. F-stat critical level (Stock and Yogo, 2003) at 10% is 16.38. All data are for 1965–1984.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

Table 8
Mechanisms: small sector, factories and wages.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Value added		Factories	Fixed capital		Manufacturing wage	
	All	Small sector		All	Small sector	Real	Nominal
	IV using groundwater × time trend						
Agricultural connections	**0.045**	**0.016**	**0.005**	**0.214**	**0.062**	**0.001**	**0.002**
	(0.016)***	(0.007)**	(0.002)**	(0.046)***	(0.006)***	(0.001)	(0.009)
Education expenditure	**0.356**	**−0.001**	**−0.003**	**1.405**	**0.198**	**0.095**	**0.019**
	(0.072)***	(0.041)	(0.012)	(0.236)***	(0.064)***	(0.050)*	(0.055)
Rural population	**−7.429**	**1.853**	**−1.730**	**−16.19**	**1.009**	**−8.879**	**−11.72**
	(4.568)	(1.765)	(1.054)*	(11.95)	(2.265)	(3.190)***	(3.283)***
Population density	**−0.785**	**0.074**	**0.189**	**2.133**	**−0.036**	**0.128**	**−0.001**
	(0.790)	(0.549)	(0.245)	(2.147)	(0.448)	(0.594)	(0.620)
Total credit	**−0.225**	**−0.078**	**−0.004**	**0.319**	**−0.143**	**0.338**	**0.291**
	(0.072)***	(0.032)**	(0.013)	(0.194)*	(0.054)***	(0.042)***	(0.042)***
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	300	195	293	280	224	300	300
First stage F-test	79.01	32.49	47.75	26.33	17.41	79.01	79.01

Standard errors in parentheses, robust to heteroskedasticity. Results are robust to different degrees of autocorrelation. All regressions include state and year fixed effects. "Agricultural connections" is the number of agricultural units connected to the electricity network per 1000 people. "Groundwater" is the proportion of state area covered with aquifers thicker than 150 mts. "Fixed capital" and "value added" are per capita for the manufacturing sector. All outcome variables are in real terms and per capita. "Small sector" are factories in the 'registered sector' with less than 50 workers using power or more than 50 and less than 100 workers without power. Variables in columns (2) to (5) have missing observations because of data was not reported in original sources for some states and years. "Wages" are log of wages for workers in the manufacturing sector. "Education expenditure" and "total credit" are in logs, per capita. "Rural population" is a share total population and, "population density" is population (in 1000 s) per square kilometer. In all cases, the coefficient of "groundwater * time trend" in the First Stage is significant at the 0.1% level. Full first stage available from the author.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

sector where the coefficient is positive and strongly significant. Its magnitude is larger than the one obtained for the whole manufacturing sector, an unsurprising result since no significant effect is present for the informal sector.[19]

Columns (3) to (8) make use of data by industry, only available from 1973 to 1984. In Columns (3) and (4) I split industries according to their intensity in electricity usage, using an input–output matrix for all India industries, and find that the effect of electrification on industrial output is greater for industries that are above the median in electricity usage.

Another way of checking whether the Green Revolution has a direct effect in the development of the industrial sector is by looking at the food processing industries. Since the Green Revolution has increased considerably food production only, it could be the case that this drives manufacturing output up by boosting the industrial processing of foodstuff. Columns (5) and (6) show that this is not the case. The coefficients for both food processing industries and the rest are very similar and, if anything, slightly larger for the non-food industries.

Finally, the same exercise could be done with respect to water use. A concern might be that states with abundant groundwater are better suited to increase their production in industries that are water intensive, such as textiles or pulp and paper.[20] If that was the case, the instrument would be affecting the dependent variable through means other than electrification. Columns (7) and (8) show positive results for both groups of industry water-intensity, that are not significantly different from each other with a larger point estimate for low-water industries. Overall, when splitting the manufacturing data according to the use of alternative inputs, only in the case of electricity I found the difference between high and low input intensity to work in the expected direction.

4.5. Mechanisms

There are direct and indirect mechanisms through which the expansion of the electricity network can affect industrialization. For example, access to electricity can lower costs, then prices and induce more consumption of manufacturing goods.[21] Murphy et al. (1989b) show that infrastructure can be an important component of a 'big push' industrialization process. By overcoming coordination problems that arise because infrastructure goods are used by many sectors at the same time, the public provision of infrastructure contributes to the development of other markets, since it has the effect of reducing the total production costs of the other sectors. In a setting with heterogeneous firms, a selection effect might be present, wherein smaller and less productive firms can break even only when electricity is available. In that sense, the effects of electrification should be present both in the intensive and extensive margins.

An alternative story could be that electrification in rural areas reduces home production time and increases labor supply (as in Dinkelman, 2010) or simply that the increase in electrification (and agricultural productivity) is freeing labor resources from the agricultural sector. That would translate into a reduction in wages that could spurt the manufacturing sector.

To test these mechanisms, I first make use of the Annual Survey of Industries' distinction of firms in the 'registered' sector between 'small sector' factories, i.e. those with less than 50 workers (and with more than 50 and less than 100 workers that do not use power), and 'census sector' factories, i.e. those with more than 100 employees. In particular, to tell apart the intensive and extensive margins of industrialization, I look at other indicators of industrial development (value added and fixed capital) for all firms and for the small sector. Columns (1), (2), (4) and (5) in Table 8 show positive and significant results. The coefficients for all firms are larger than for the small

[19] Unreported results using census data for two years also show that employment in the rural manufacturing sector also goes up with the instrumented measure of electrification. Overall, there is no enough data on the rural manufacturing sector to produce further checks.

[20] However, the effect could go the other way. For example, Keskin (2009) shows that industries and the agricultural sector might compete for water and that industrialization might hurt farmers.

[21] A look at the input–output coefficient matrix for industries in the early 1990s India shows that for many industries, such as heavy chemicals, cement or non-ferrous metal, electricity is more than 10% of their total costs and can be as high as 17% (Ministry of Industry, Government of India, 1993).

Table 9
Alternative measures of groundwater.

	(1)	(2)	(3)	(4)	(5)
	Log manufacturing output				
	IV using groundwater × time trend			Alternative first stage	
Agricultural connections	**0.030**	**0.032**	**0.035**	**0.027**	**0.025**
	(0.008)***	(0.008)***	(0.013)***	(0.007)***	(0.007)***
Education expenditure	**0.166**	**0.165**	**0.164**	**0.166**	**0.167**
	(0.067)**	(0.068)**	(0.072)**	(0.065)**	(0.064)***
Rural population	**−6.125**	**−5.779**	**−5.077**	**−7.007**	**−7.341**
	(3.761)*	(3.691)	(4.455)	(3.721)*	(3.681)**
Population density	**1.134**	**1.119**	**1.089**	**1.172**	**1.186**
	(0.642)*	(0.639)*	(0.147)	(0.645)*	(0.643)*
Total credit	**0.023**	**0.022**	**0.021**	**0.024**	**0.025**
	(0.044)	(0.044)	(0.044)	(0.044)	(0.045)
Specification	>100 m thick	All aquifers	Quantity of groundwater (2004)	Square	Flexible
Observations	300	300	300	300	300
First stage					
	Agricultural connections				
Groundwater × time trend	**0.594**	**0.627**	**1.341**	**1.142**	See Fig. 4
	(0.096)***	(0.094)***	(0.337)***	(0.333)***	
Groundwater × time squared				**−0.019**	
				(0.016)	
Education expenditure	**−0.499**	**−0.131**	**−0.252**	**−0.995**	**−1.090**
	(1.061)	(0.905)	(0.163)	(0.875)	(0.807)
Rural population	**−222.3**	**−206.9**	**−241.74**	**−220.1**	**−225.90**
	(20.12)***	(18.27)***	(22.85)***	(19.95)**	(20.93)***
Population density	**−19.05**	**−29.18**	**−22.35**	**−13.74**	**−12.59**
	(5.909)***	(5.722)***	(9.198)**	(6.054)**	(6.168)**
Total credit	**−0.494**	**−0.928**	**0.087**	**−0.356**	**−0.352**
	(0.436)	(0.419)**	(0.592)*	(0.426)	(0.444)
First stage F-test	38.27	44.40	15.79	43.85	6.06
R-square	0.94	0.94	0.93	0.95	0.96

Standard errors in parentheses, robust to heteroskedasticity. Results are robust to different degrees of autocorrelation. "Agricultural connections" is the number of agricultural units connected to the electricity network per 1000 people. "Log manufacturing output" is the log of real per capita manufacturing output. In column (1), "groundwater" is the proportion of state area covered with aquifers thicker than 100 mts, and in column (2), is the proportion of state area covered with aquifers with any positive groundwater depth. Column (3) uses the net groundwater available (in cubic meters per squared km), as estimated by Central Ground Water Board (2006). In columns (4) and (5) "groundwater" is the proportion of state area covered with aquifers thicker than 150 mts. "Education expenditure" and "total credit" are in logs, per capita. "Rural population" is a share total population and, "population density" is population (in 1000 s) per square kilometer. The "flexible" specification uses "groundwater × year dummies" as an instrument. The coefficients for the first stage are shown in Fig. 4 and available from the author. All regressions use state and year fixed effects.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

sector, suggesting that the effect is not all coming from smaller firms.[22] Another way of testing the extensive margin is by looking at whether new factories open. In column (3) I find that electrification is significantly associated with more factories. The magnitude suggests that 1 extra electricity connection per 1000 people would be associated with 0.005 factories per 1000 people. For a state with the average population of 1965, i.e. 30 million, that would mean 150 extra factories. These results provide evidence of effects happening both at the intensive and the extensive margins: a deepening of the industrial sector among big and small firms and the presence of a selection effect where new and smaller firms also benefit by the expansion of the electricity network. However, the marginal effects are significantly lower for the smaller sector. This indicates that the most of the increase in manufacturing output might come from the intensive margin expansion.

To test the indirect channel, I use information on real and nominal wages in the manufacturing sector. In both cases, shown in columns (6) and (7), there is no evidence that wages have dropped significantly in states that electrified more. This is not consistent with the idea that manufacturing output has increased because of a drastic reduction in wages (that would have followed an stark increase in labor availability).

[22] This holds when restricting the dataset to state-year observations that are both available for all firms and for the small sector.

5. Robustness checks

In this section I investigate further to what extent the main set of results presented in Section 3.2 hold when using alternative measures of groundwater availability and electrification.

5.1. Other measures of groundwater

The evidence presented so far has linked a measure of groundwater availability that only considered the presence of aquifers deeper than 150 mts. The district level evidence in Fig. 6 suggested that this measure was a good predictor of divergence in pumpset use. What happens with the main results when I relax the constraint imposed on the measure of groundwater availability?

I consider three measures to check the robustness of results in Table 4. In the first and second measures I use, respectively, the proportion of state area with aquifers deeper than 100 mts and the proportion of districts with any positive groundwater depth. Finally, the third alternative measure uses data on net groundwater availability (in cubic meters per squared km), as estimated by Central Ground Water Board (2006) for the year 2004. Table 9, columns (1) to (3) show results when instrumenting with the interaction of the different groundwater measures and a linear time trend. Across specifications the second stage shows a positive and significant effect of electrification on manufacturing outcomes and strong first stage results.

Table 10
Alternative measures of electrification.

	(1)	(2)	(3)	(4)	(5)
	Log manufacturing output				
	IV using groundwater × time trend				
Percentage villages electrified	**0.998**				
	(0.296)**				
Connected load (Kwpc)		**0.003**			
		(0.0009)***			
Industrial connections (per 1000 people)			**0.353**		
			(0.142)**		
Generation capacity (Kwpc)				**0.010**	
				(0.004)**	
T&D losses					**−0.711**
					(0.510)
Education expenditure	**0.038**	**0.153**	**0.190**	**0.032**	**0.232**
	(0.085)	(0.067)**	(0.081)**	(0.102)	(0.202)
Rural population	**−10.17**	**−6.872**	**−2.688**	**−3.243**	**−10.91**
	(3.225)***	(4.100)*	(5.916)	(5.627)	(6.759)
Population density	**2.447**	**1.577**	**1.245**	**1.219**	**−3.526**
	(0.709)***	(0.651)**	(0.642)*	(0.618)**	(3.684)
Total credit	**0.091**	**0.064**	**0.137**	**0.092**	**−0.043**
	(0.057)	(0.047)	(0.067)**	(0.055)*	(0.133)
Observations	300	300	300	300	300
First stage					
Groundwater × time trend	**0.020**	**5.99**	**0.056**	**1.917**	**−0.003**
	(0.005)***	(0.930)***	(0.017)***	(0.536)***	(0.002)
Education expenditure	**0.106**	**0.165**	**−0.130**	**10.91**	**0.007**
	(0.056)*	(0.068)**	(0.173)	(4.206)***	(0.030)
Rural population	**−2.589**	**−5.779**	**−28.55**	**−922.9**	**0.290**
	(1.255)**	(3.691)	(4.591)***	(99.81)***	(0.698)
Population density	**−1.688**	**1.119**	**−1.373**	**−44.39**	**−0.593**
	(0.291)***	(0.639)*	(1.360)	(44.33)	(0.358)
Total credit	**−0.076**	**0.022**	**−0.011**	**−7.462**	**−0.008**
	(0.031)**	(0.044)	(0.099)***	(2.320)***	(0.016)
First stage F-test	15.84	41.42	11.29	12.78	1.71
R-square	0.95	0.94	0.94	0.92	0.17

Standard errors in parentheses, robust to heteroskedasticity. Results are robust to different degrees of autocorrelation. "Log manufacturing output" is the log of real per capita manufacturing output. "Groundwater" is the proportion of state area covered with aquifers thicker than 150 mts. "Connected load" and "generation capacity" are measured in KW per million people. "Industrial connections" is measured per 1000 people. "T&D losses" are Transmission and Distribution loses as a proportion of total electricity produced. "Education expenditure" and "total credit" are in logs, per capita. "Rural population" is a share total population and, "population density" is population (in 1000 s) per square kilometer. All regressions use state and year fixed effects.

*** Significant at 1%.
** Significant at 5%.
* Significant at 10%.

In columns (4) and (5) I present results using the initial measure of groundwater in different first stage specifications, namely a quadratic function and a fully flexible (as in Eq. (2), respectively. In column (4) the quadratic term in the first stage is not significant, even though overall the strategy remains strong.[23] Similarly, Fig. 4 shows that the coefficients after 1968 are significantly different from 0, but the first stage is overall weaker. This suggests that the use of the linear time specification is the better fit for the purposes of this investigation.

5.2. Other measures of electrification

The use of alternative measures of electrification provides a similar picture. I use information on the percentage of villages electrified, on the state technical capacity both to provide (i.e. connected load) and to generate electricity and on the number of industrial connections. Finally, it might be the case that the expansion of the network has also brought a reduction or an increase in quality that might explain part of the results. To test this I use information on Transmission and Distribution losses (as a percentage of total electricity available).

Results in Table 10 show that the link between electrification and manufacturing output holds for the four alternative measures of network expansion. The first stage statistics are lower than for the measure of agricultural connections, suggesting the stronger link between groundwater and rural electrification.[24] The result on the quality measure, in column (5), does not provide evidence of a positive or negative link with manufacturing output.

There is no consistent information on electricity prices by type of consumers for the period analyzed, but the anecdotal evidence (see Tongia, 2003, for example) suggest a significant cross-subsidy to rural and household consumption from industrial and commercial users across most Indian states. However, I can use information on average electricity tariffs and average cost, provided from 1974 by the Planning Commission (2001). When running the instrumented measure of agricultural electricity connection on the ratio average tariff-average cost I find that states that have electrified more have, on average, a 1.1 percentage point lower ratio. That means that pricing policies seem to be less sustainable in states with more rural electrification. Furthermore, a regression of the tariff-cost ratio on manufacturing output shows a negative and significant

[23] A similar pattern emerges for a cubic first stage specification.

[24] The variation in rural electrification explained by the time-varying effects of groundwater would, however, be a very strong predictor of the other measures of electrification, such as industrial connections.

coefficient. That means that states with more sustainable tariffs (and probably less cross-subsidies to agricultural and household consumers) have observed a lower industrial growth. Overall, this would suggest that, if these higher prices of electricity for industrial users affect significantly manufacturers' outcomes, the main results would underestimate the effects of network expansion in a setting without cross-subsidies.

6. Conclusion

The role that infrastructure provision plays in improving economic and social outcomes in developing countries is part of the conventional wisdom among academics, policy makers and the population at large. Despite this consensus, the academic literature has not provided a substantial amount of empirical investigations aimed at understanding and quantifying the effects of large scale infrastructure provision. The challenge of this paper is to document the effect of electrification in industrial output by looking at a panel of Indian states from 1965 to 1984. To do so, I address the endogenous placement of infrastructure by looking at the time-varying effect of time-invariant geographic characteristics on electrification after the start of the Green Revolution in India. The need for timely irrigation that could be cheaply supplied by pumping water from the water table using electrified pumpsets generated a growing demand for electricity and subsequent expansion of the electricity grid in some states. Instrumental variables estimations show that one standard deviation difference in electrification can explain around 14% of the differential level in manufacturing output across Indian states. The expansion of the network also helped the entry and performance of smaller firms. These results are robust to a series of tests regarding the power and validity of the instruments and support the idea that the lack of infrastructure is a major constraint on economic activity in developing countries. The magnitude of the results suggests that expanding the electricity network – including rural areas – should be considered seriously as a policy option for promoting industrial development. Still, a degree of caution is needed here: electrification might have had these effects because it was a significant binding constraint for manufacturing firms across Indian states. However, this might not always be the case, in particular if the population is subject to other constraints, such as transport infrastructure or credit availability that would make the provision of electricity less effective.

Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.jdeveco.2011.06.010.

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